8 Sampling

Objective

After completing this chapter, user will be able to understand

1. Sampling theorem for low pass signals
2. Sampling techniques
3. Sampling of sinusoidal signals
4. Sampling theorem for band-pass signals

Introduction

1. A continuous-time signal can be completely represented by and recoverable from knowledge of its values or samples at points equally spaced in time.
2. Under certain conditions, a continuous-time signal can be completely recovered from a sequence of its samples provide a mechanism for representing a continuous-time signal by a discrete time signal.
3. Sampling is also frequently performed on discrete-time signals to change the effective data rate; an operation termed subsampling.
4. In this case, the sampling process discards certain values of the signal.

Sampling

Types of message signals:

1. Baseband signal
2. Bandpass signal
3. Sinusoidal signal

Baseband Signal

1. It is a naturally existing signal.
2. Lower cutoff frequency very close to 0 Hz.

Example:

Voice signal: 300Hz – 3.5KHz

Audio signal: 20Hz – 20KHz

Video signal: 0Hz – 4.5MHz

Spectrum:

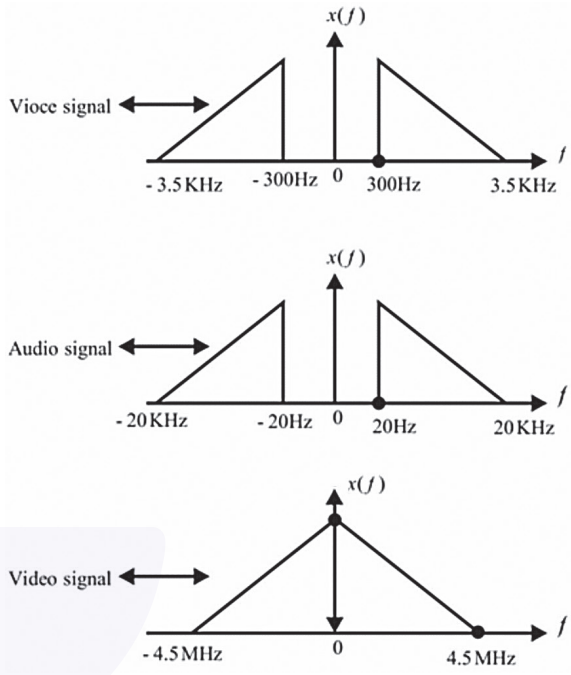


Fig. 8.1

Note:

Baseband signals: Lowpass signal.

Example:

Video signal.

Bandpass Signal

1. These are fabricated signal.
2. These signals have lower cutoff frequencies very much greater than 0Hz (MHZ).

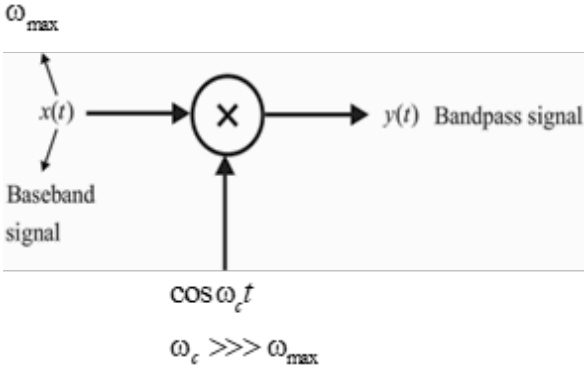


Fig. 8.2

$m(t)$[video] $\longleftrightarrow$

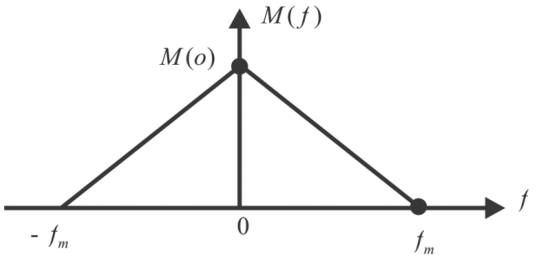


Fig. 8.3

$s(t) = m(t)\cos 2\pi f_z t \longleftrightarrow$

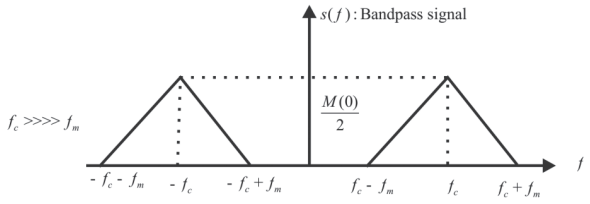


Fig. 8.4

1. m(t): Voice signal:

$s(t) = m(t)\cos(2\pi \times 10^6 t)$: Plot spectrum of s(f).

Solution:

$m(t) \longleftrightarrow$

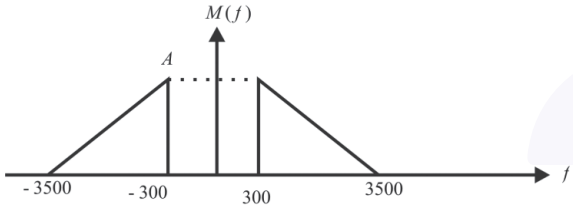


Fig. 8.5

$s(t) = m(t)\cos\left(2\pi \times 10^6 t\right)$

$f_c = 1000KHz$

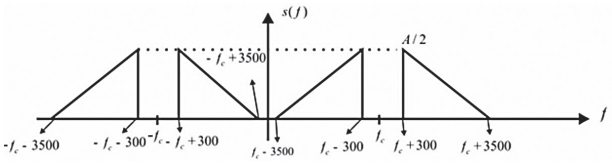


Fig. 8.6

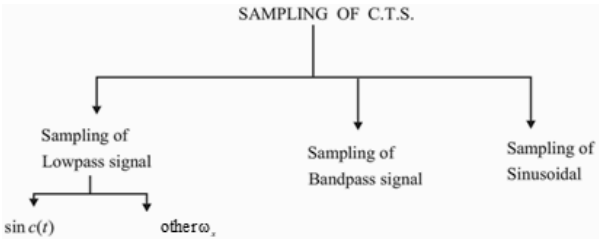
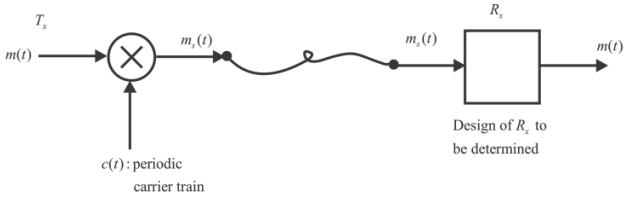


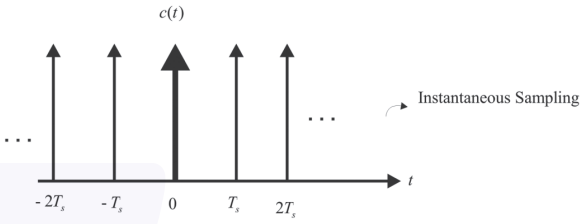
Fig. 8.7

Sampling of Low-Pass Signal

1. Sampling is a process of converting continuous time signals into discrete-time signals.
2. Sampling is useful only when continuous time signal is recovered from discrete-time signal.

Setup:





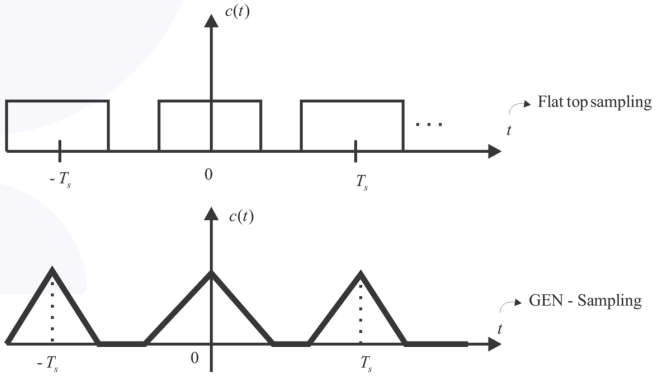


Fig. 8.8

Instantaneous Sampling

1. $m(t) \rightarrow$ Lowpass $\overset{F.T.}{\longleftrightarrow}$

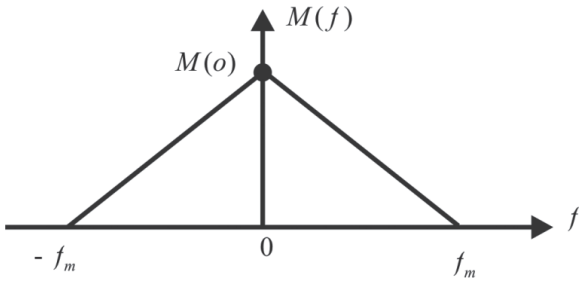


Fig. 8.9

$m(t)$ is Bandlimited to f_m Hz

max freq component of $m(t) = f_m Hz$

2.

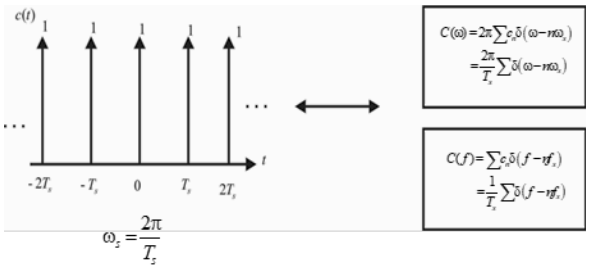


Fig. 8.10

SETUP

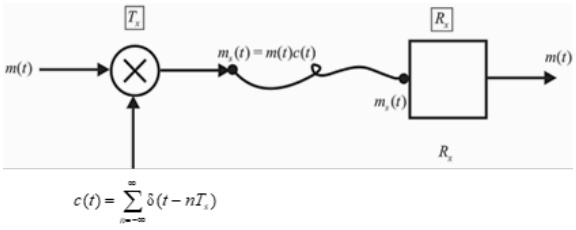


Fig. 8.11

Time Domain Analysis

at Tx : $m_s(t) = m(t).c(t)$

$$m_s(t) = m(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)$$

$$m_s(t) = \sum_{n=-\infty}^{\infty} m(t)\underbrace{\delta(t-nT_s)}_{t=nT_s}$$

$$= \sum_{n=-\infty}^{\infty} m(nT_s)\delta(t-nT_s)$$

$$m_s(t) = +.....m(-T_s)\delta(t+T_s) + \underbrace{m(0)}\,\delta(t) + m(T_s)\delta(t-T_s) + ...$$

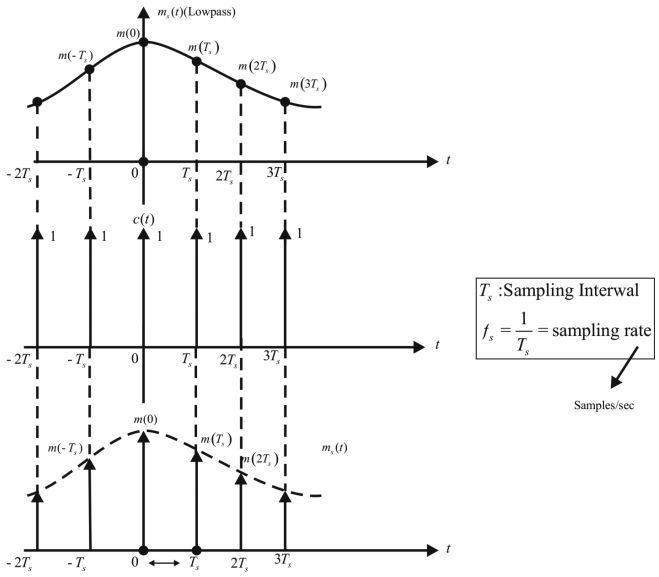


Fig. 8.12

at R_x :

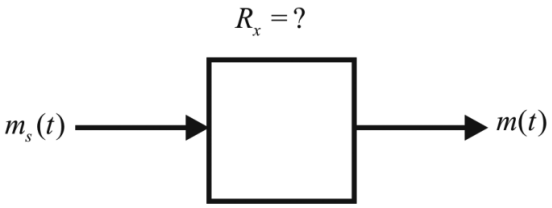


Fig. 8.13

Frequency Domain Analysis

Case 1:

$$m(t) \longleftrightarrow \boxed{M(\omega)}$$

$$c(t) \longleftrightarrow c(\omega) = {}^{2\pi}\!/_{Ts}\sum_{n=-\infty}^{\infty}\delta(\omega - n\omega_s)$$

$$m_s(t) = m(t)c(t) \longleftrightarrow M_s(\omega) = \frac{1}{2\pi}[M(\omega) * C(\omega)]$$

$$M_s(\omega) = \frac{1}{2\pi}\left[M(\omega) * \frac{2\pi}{Ts}\sum_{n=-\infty}^{\infty}\delta(\omega - n\omega_s)\right]$$

$$M_s(\omega) = f_s\left[M(\omega) * \sum\delta(\omega - n\omega_s)\right]$$

$$M_s(\omega) = f_s\left[\sum_{n=-\infty}^{\infty} M(\omega) * \delta(\omega - n\omega_s)\right]$$

$$M_s(\omega) = f_s\sum_{n=-\infty}^{\infty}[M(\omega - n\omega_s)]$$

$$M_s(\omega) = f_s[..... \underbrace{M(\omega + \omega_s)}_{clones} + \underbrace{M(\omega)}$$

$$+ \underbrace{M(\omega - \omega_s)}_{clones} + ...]$$

Case 2:

$$m(t) \longleftrightarrow M(f)$$

$$c(t) \longleftrightarrow c(f) = \sum_{n=-\infty}^{\infty} c_n\delta(f - nf_s)$$

$$c(f) = f_s\sum_{n=-\infty}^{\infty}\delta(f - nf_s)$$

$$m_s(t) = m(t)c(t) \longleftrightarrow M_s(f) = M(f) * c(f)$$

$$M_s(f) = M(f) * f_s\sum\delta(f - nf_s)$$

$$M_s(f) = f_s\left[M(f) * \sum\delta(f - nf_s)\right]$$

$$\boxed{M_s(f) = f_s\sum_{n=-\infty}^{\infty} M(f - nf_s)}$$

$$M_s(f) = f_s[..... \underbrace{M(f + f_s)}_{clones} + \underbrace{M(f)} + \underbrace{M(f - f_s)}_{clones} + ...]$$

Conclusions:

$x(t) \longleftrightarrow x(\omega)$

$x(t) \cdot \sum \delta(t - nT_s) \longleftrightarrow f_s \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$

$x(t) \longleftrightarrow x(f)$

$x(t) . \sum \delta(t - nT_s) \longleftrightarrow f_s \sum_{n=-\infty}^{\infty} X(f - nf_s)$

Spectral Analysis

$m(t) \longleftrightarrow$

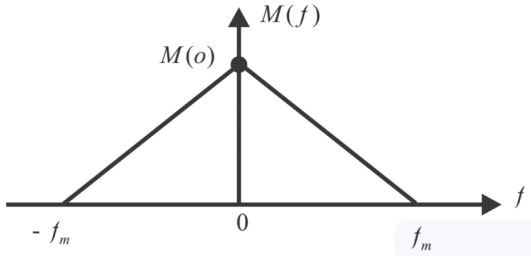


Fig. 8.14

Case 1:

$f_s > 2f_m \rightarrow f_s - f_m > f_m$

T_x

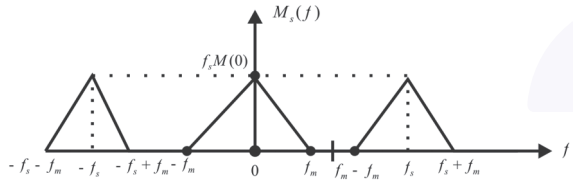


Fig. 8.15

R_x :

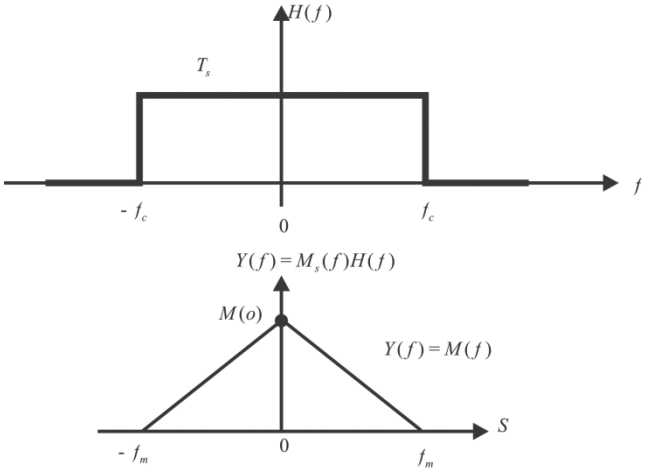


Fig. 8.16

Conclusion:

Tx:

1. $f_s >> 2f_m \rightarrow$ No overlapping between adjacent sidebands.
   $\rightarrow$ No Aliasing

Rx:

2. Filter: $f_m < f_c < f_s - f_m \rightarrow$ certain frequency will be passed.
3. For perfect recovery:
   $(f_s >> 2f_m) \cap (f_m < f_c < f_s - f_m)$

PBG	y(t)
1	$f_s m(t)$
T_s	$m(t)$
K	$Kf_s m(t)$

Table 8.1

Case 2:

$f_s = 2f_m$: $f_s - f_m = f_m$

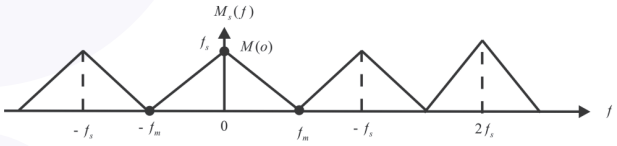


Fig. 8.17

R_x

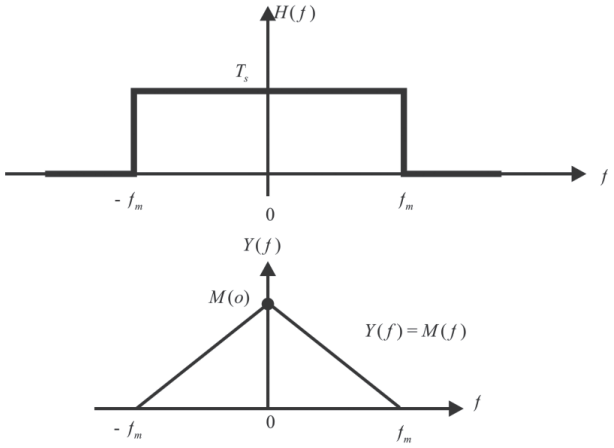


Fig. 8.18

Conclusion:

For perfect recovery:

1. $(f_s = 2f_m) \cap (Ideal\ LPF\ with\ f_c = 2f_m)$
2. Pass band gain of LPF = T_s

Case 3:

$f_s < 2f_m$: $f_s - f_m < f_m$.

Tx:

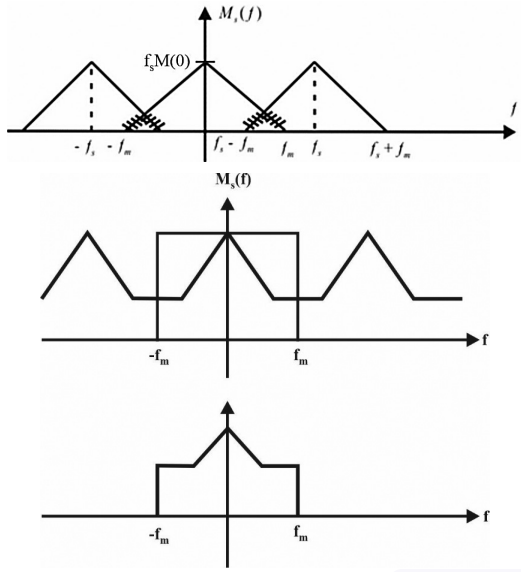


Fig. 8.19

Conclusion:

$f_s < 2f_m$: No recovery or reconstruction is possible.

Test Your Understanding

1. The amplitude spectrum of the sampled signal of a signal $x(t)$ is shown in the figure below.

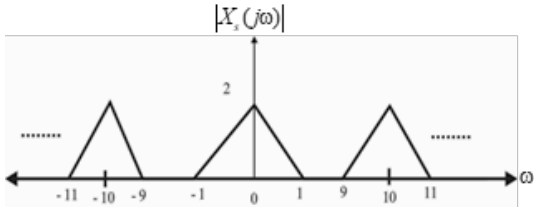


Fig. 8.20

The amplitude spectrum of original signal $x(t)$ will be

(A)

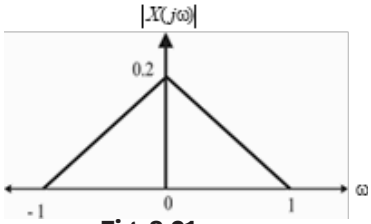
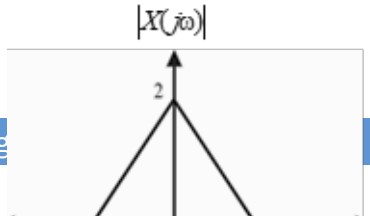


Fig. 8.21



(B)

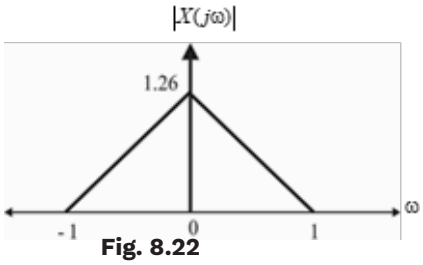


Fig. 8.22

(C)

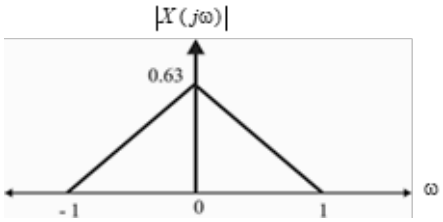


Fig. 8.23

Solution: (C)

Lowpass Sampling Theorem

A lowpass signal band limited to f_m *Hz* can be sampled and reconstructed from it samples only if:

$$[f_s \underset{\text{Tx}}{>>} 2f_m] \cap (f_m < f_c < f_s - f_m)_{\text{Rx(filter)}} \cap (PBG = T_s)$$

Nomenclature:

$f_s >>> 2f_m$: Over sampling

$f_s = 2f_m$: Critial sampling

$f_s << 2f_m$: Under sampling

Desired:

→ Sampling : Over sampling

Modulation : Under modulation

Important Terms

1. **Nyquist rate**: "Minimum sampling rate for lowpass signal."

$f_s > 2f_m$

$(f_s)_{\min} = 2f_m \rightarrow f_{NY}$

2. **Nyquist interval:**

$$T_{NY} = \frac{1}{f_{NY}}$$

$m(t) \longleftrightarrow$

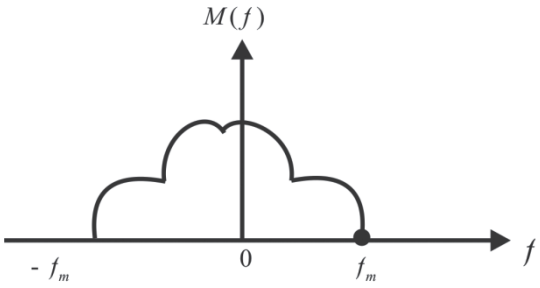


Fig. 8.24

1. $m(t)$ → $f_{max} = f_m$, $f_N = 2f_m$

2. $m(-t)$ → $f_{max} = f_m$, $f_N = 2f_m$

3. $m(t - t_0)$ → $f_{max} = f_m$, $f_N = 2f_m$

4. $m(at)$ → $f_{max} = af_m$, $f_N = 2(af_m)$

5. $m(-at + b)$ → $f_{max} = af_m$, $f_N = 2(af_m)$

6. $\frac{dm(t)}{at}$ → $f_{max} = f_m$, $f_N = 2f_m$

7. $\int_{-\infty}^{t} m(\varepsilon)d\varepsilon$ → $f_{max} = f_m$, $f_N = 2f_m$

8. $m^2(t)$ → $f_{max} = 2f_m$, $f_N = 2\{2f_m\}$

9. $m^K(t)$ → $f_{max} = Kf_m$, $f_N = 2(Kf_m)$

10. $m(t)\cos 2\pi f_c t$ → $f_{max} = f_c + f_m$, $f_{NY} = 2(f_c + f_m)$

Combination of Signals

$x_1(t) \longleftrightarrow$

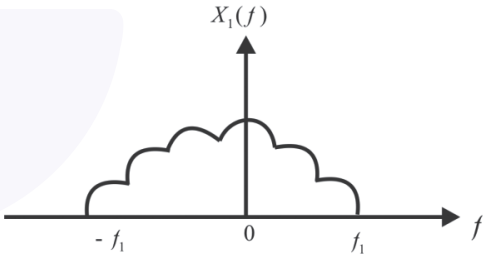


Fig. 8.25

$x_2(t) \longleftrightarrow$

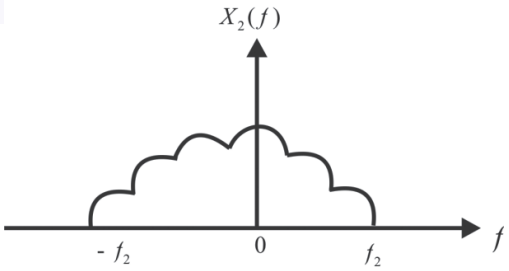


Fig. 8.26

$x_1(t)$ → $f_{max} = f_1$, $f_{N_1} = 2f_1$

$x_2(t)$ → $f_{max} = f_2$, $f_{N_2} = 2f_2$

(1) $x_1(t) \pm x_2(t)$ → $f_{max} = \max(f_1, f_2)$, $f_N = \max(2f_1, 2f_2) = \max(f_{N_1}, f_{N_2})$

(2) $x(t) = x_1(t)x_2(t)$ → $f_{max} = (f_1 + f_2)$, $f_N = 2f_1 + 2f_2 = f_{N_1} + f_{N_2}$

(3) $x(t) = x_1(t) * x_2(t)$ → $f_{max} = \min(f_1, f_2)$, $f_N = \min(2f_1 + 2f_2) = \min(f_{N_1} + f_{N_2})$

Points to Remember

M(t)	NR
$\sin c(t)$	1 Hz
$\sin c(at)$	a Hz
$\sin c^k(at)$	ka Hz
$\sin c(-at + b)$	ka Hz
$d / dt(\sin c(t))$	1 Hz
$\int \sin c(t)dt$	1 Hz
$\sin c(at) + \sin c(bt)$	max (a,b) Hz
$\sin c(at) \times \sin c(bt)$	(a+b) Hz
$\sin c(at) * \sin c(bt)$	min (a, b) Hz

Table 8.2

1.

$$x(t) = \left[\left\{\frac{\sin^3(2t)}{3t^3}\right\} * \left\{\int_{-\infty}^{t} \frac{\sin^2 5tdt}{3t^2}\right\}\right] \times \left[\left(\frac{\sin 3\pi t}{t}\right) + \frac{d}{dt}\left(\frac{\sin^3 3t}{t^3}\right)\right]$$

Solution:

$$x(t) = \left[\left\{\frac{\sin^3(2t)}{3t^3}\right\} * \left\{\int_{-\infty}^{t} \frac{\sin^2 5tdt}{3t^2}\right\}\right] \times \left[\left(\frac{\sin 3\pi t}{t}\right) + \frac{d}{dt}\left(\frac{\sin^3 3t}{t^3}\right)\right]$$

$\frac{6}{\pi}$ $\frac{10}{\pi}$ ↓min $\frac{6}{\pi}$

3 $\frac{9}{\pi}$ ↓max 3

$$f_N = (3 + \frac{6}{\pi})Hz$$

Sampling of Signals using General Pulses

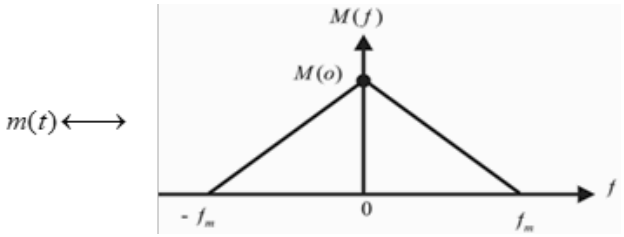


Fig. 8.27

T_s $\quad c(t) \longleftrightarrow c(f) = \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_s)$

$$m_s(t) \longleftrightarrow M_s(F) = \sum_{n=-\infty}^{n=\infty} c_n M(f - nf_s)$$

T_x $\quad M_s(f) = ...C_{-1}M(f + f_s) + \underline{C_o M(f)} + C_1(f - f_s) + ...$

$f_s >> 2fm :\rightarrow$ Clones will be well separated.

Rx:

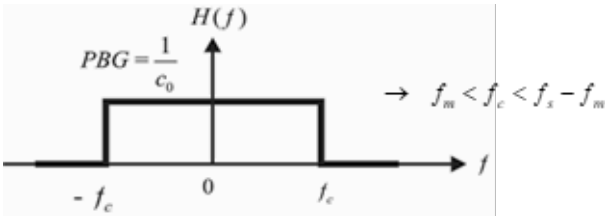


Fig. 8.28

Case 1:

Let c(t)is rectangular pulse train

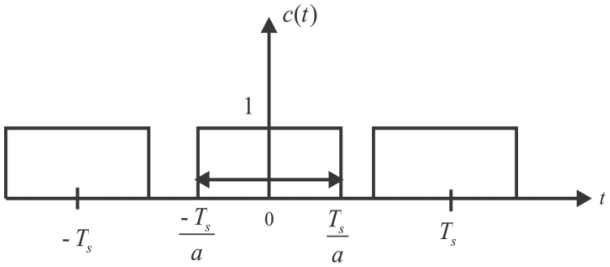


Fig. 8.29

$$\begin{cases} C_0 = \dfrac{2}{a} \\ C_n = C_0 \,\mathrm{Sin\,c}\left[n\left(\dfrac{2T_s}{aT_s}\right)\right] = C_0 \sin\left[\dfrac{2n}{a}\right] \end{cases}$$

$M(t) \longleftrightarrow M(f) =$

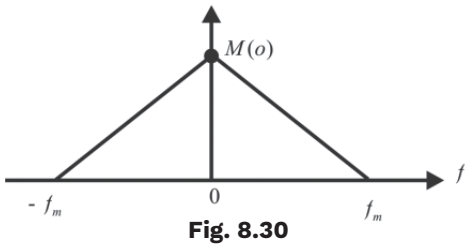


Fig. 8.30

$c(t) \longleftrightarrow c(f)$

$$m_s(t) \longleftrightarrow \sum_{n=-\infty}^{\infty} C_n M(f - nf_s) = M_s(f)$$

$$M_s(f) = +.....C_{-1}M(f + f_s) + C_0M(f) + C_1M(f - f_s) + ...$$

Tx.

$f_s > 2f_m$

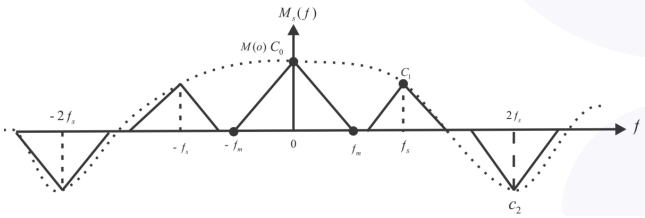
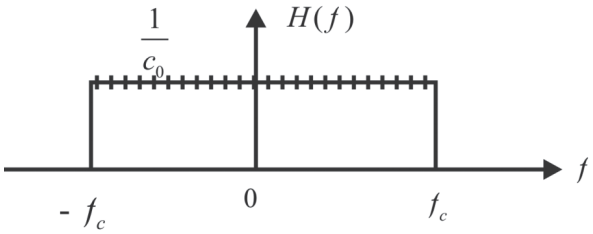


Fig. 8.31

Rx.



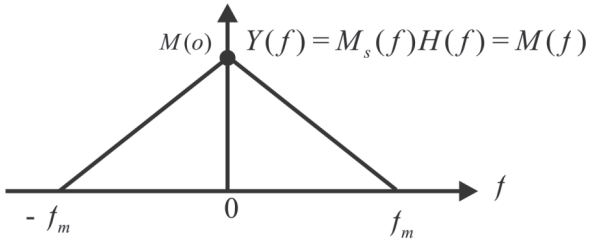


Fig. 8.32

Perfect recovery:

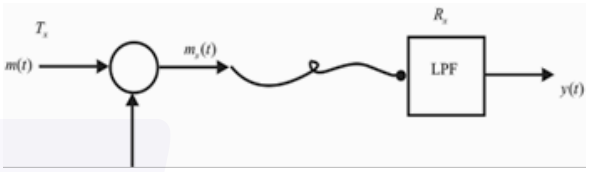
$$(f_s > 2f_m) \cap (f_m < f_c < f_s - f_m) \cap \left(PBG = \frac{1}{6}\right)$$

PBG	G(t)
1	$c_o m(t)$
$\frac{1}{c_o}$	M(t)
K	$kc_o m(t)$

Table 8.3

Sampling of Sinusoidal Signals

Instantaneous sampling: $m(t) = \cos 2\pi f_m t$



$$\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longleftrightarrow c_n = \frac{1}{T_s}$$

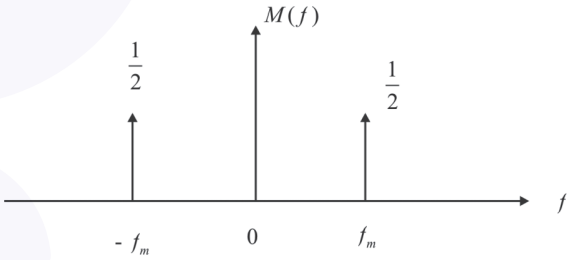


Fig. 8.33

$c(t) \longleftrightarrow c(f)$

$$M_s(t) \longleftrightarrow M_s(f) = \sum_{n=-\infty}^{\infty} C_n M(f - nf_s)$$

$$= f_s \sum_{n=-\infty}^{\infty} M(f - nf)$$

$$f_s[.....M(f + f_s) + M(f) + M(f - f_s) + ...]$$

Case 1:

$f_s > 2f_m : \; f_s - f_m > f_m$

Tx:

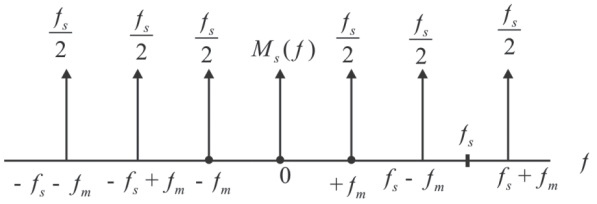


Fig. 8.34

Rx:

Perfect recovery:

1. $(f_s > 2f_m) \cap (f_m < f_c < f_s - f_m) \cap (PBG = \text{const} : |f| \le f_m)$
2. $(f_s > 2f_m) \cap (\text{proper BPF}) \cap \begin{pmatrix} \text{with PBG = constant in} \\ \text{respective range} \end{pmatrix}$

Case 2:

$f_s = 2f_m \quad f_s - f_m = f_m$

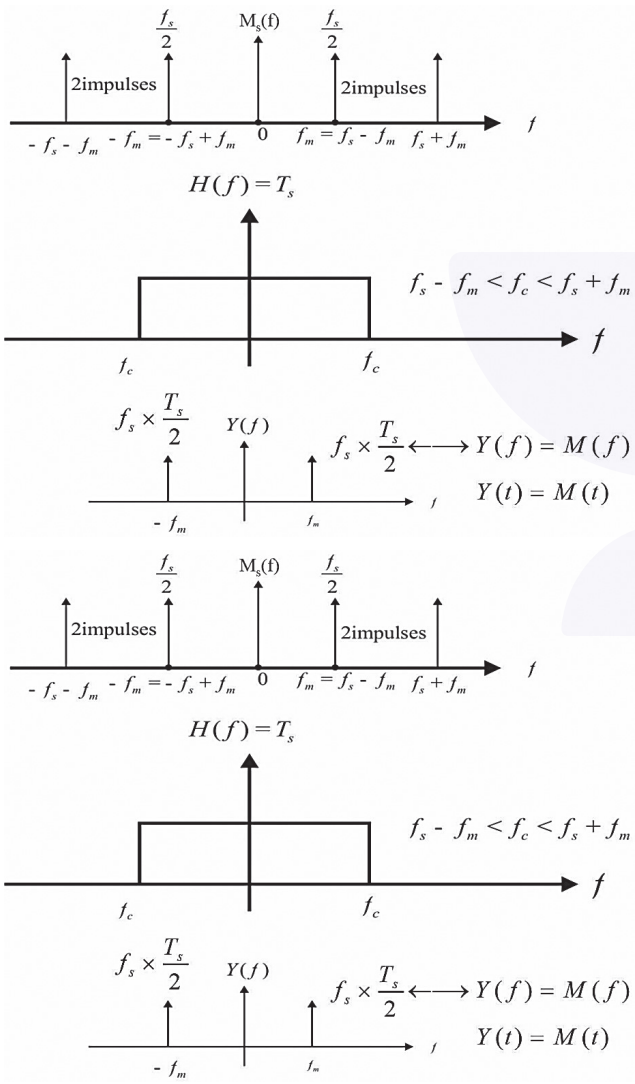


Fig. 8.35

Perfect recovery:

$(f_s = 2f_m) \cap (f_m < f_c < f_s + f_m) \cap (PGB = T/2_s : |f| \le f_m)$

Case 3:

$f_s < 2f_m = f_s - f_m < f_m$

Tx:

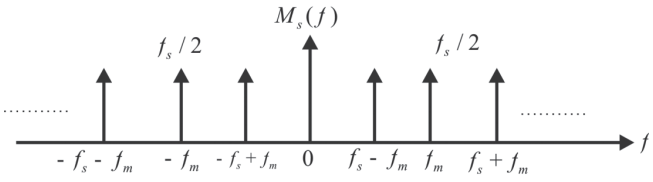


Fig. 8.36

Rx:

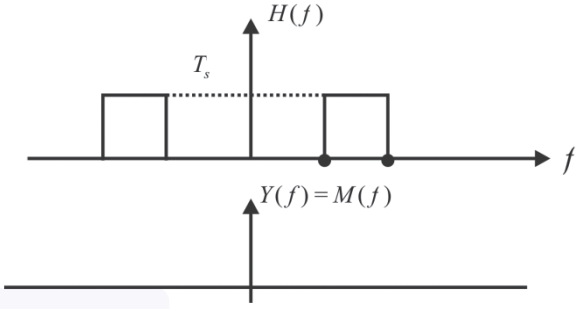


Fig. 8.37

Case 4:

$f_s < f_m \quad f_s - f_m < 0$

$-f_s + f_m > 0$

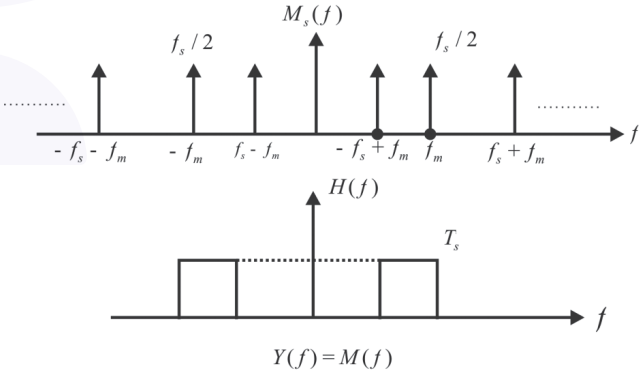


Fig. 8.38

Points to remember:

Conclusion:

Recovery chart →

	LPF	BPF	PBG
$f_s > 2f_m$	✓	✓	T_s
$f_s = 2f_m$	✓	✓	$T_s/2$
$f_s < 2f_m$	×	✓	T_s
$f_s < f_m$	×	✓	T_s

Calculation of Frequency in Sampled Signal

1. $m(t) = \cos 2\pi f_m t$ or $\sin 2\pi f_m t \longrightarrow f_m$

$$c(t) = \Sigma\delta(t - nT_s) \longrightarrow 0, f_s, 2f_s, 3f_s, 4f_s \ldots\ldots$$

$$m_s(t) = m(t).c(t) \longrightarrow 0 \pm f_m \begin{matrix} \nearrow 0 + f_m \\ \searrow |0 - f_m| \end{matrix}$$

$$f_s \pm f_m \begin{matrix} \nearrow f_s + f_m \\ \searrow |f_s - f_m| \end{matrix}$$

$$2f_s \pm f_m \begin{matrix} \nearrow 2f_s + f_m \\ \searrow |2f_s - f_m| \end{matrix}$$

2. $m(t) = \cos 2\pi f_1 + + \sin 2\pi f_2 t \longrightarrow f_1, f_2$

$$c(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) \longrightarrow 0, f_s, 2f_s, 3f_s \ldots\ldots$$

$0 \pm f_1$	$0 \pm f_2$
$f_s \pm f_1$	$f_s \pm f_2$
$2f_s \pm f_1$	$2f_s \pm f_2$

3. $m(t) = \cos 2\pi f_m t \longrightarrow f_m$

$$m_s(t) \longrightarrow 0 \pm f_m$$
$$f_s \pm f_m$$
$$2f_s \pm f_m$$
$$4f_s \pm f_m$$

4. $m(t) = \cos 2\pi f_m t \longrightarrow f_m$

$$m_s(t) \longrightarrow 0 \pm f_m \quad 3f_s \pm f_m$$
$$f_s \pm f_m \quad 5f_s \pm f_m$$
$$2f_s \pm f_m \quad \ldots\ldots\ldots\ldots$$

Summary

Lowpass signals ⟶ N.R. and N.I. for SINC
⟶ Filtering Based

Sinusodal signals ⟶ Filtering Based Frequency Indentification

1. $x(t) = u(t + T_0) - u(t - T_0)$

Signal $x(t)$ can undergo impulse train sampling without aliasing provided sampling period $T_s < 2T_0$.

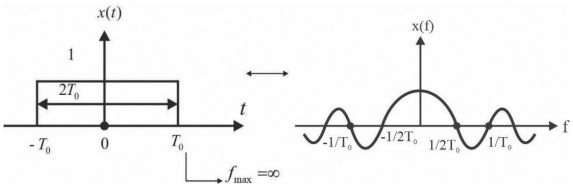


Fig. 8.39

Solution:

$$\boxed{\text{zero cross frequency} = \left(\frac{1}{2T_0}\right)}$$

$f_s > 2f_{max} \rightarrow$ Not possible

Statement is wrong.

2. Signal g(t) can undergo impulse Train sampling without aliasing provided sampling period $Ts < \pi / \omega_0$

$$G(\omega) = \mho\ (\omega + \omega_0) - \mho\ (\omega - \omega_0)$$

Solution:

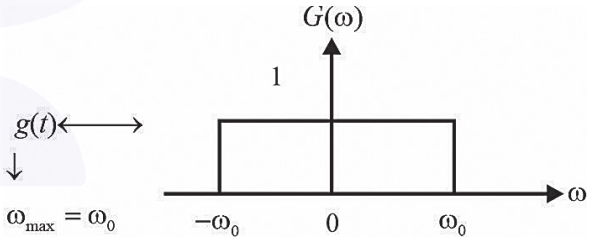


Fig. 8.40

$$\omega_s \geq 2\omega_0$$

$$\frac{2\pi}{T_s} \geq 2\omega_0$$

$$T_s \leq \frac{\pi}{\omega_0}$$

Statement is correct.

Test Your Understanding

1. For a signal $x(t)$, the Nyquist rate is 100 kHz. Match List I (signal) with List II (Nyquist rate) and choose the correct answer using the codes given below

List I (Signal)	**List II (Nyquist rate)**
P. $\frac{dx(t)}{dt}$	1. 50 kHz
Q. $x^2(t)$	2. 100 kHz
R. $x(t)\cos(300\pi t)$	3. 200 Hz
S. $x(2t)$	4. 400 Hz

Codes:

	P	Q	R	S
(A)	2	3	4	1
(B)	1	4	2	3
(C)	3	2	1	4
(D)	1	2	4	3

Solution: (A)

2. $x(t) = 5\times10^{-6}m(t) \quad \sum_{n=-\infty}^{\infty}\delta(t-nT_s)$

$p(t) = 5\times10^{-6}m(t)$

$p(t) = 5\times10^{-5}\cos(8\pi\times10^3 t)$

$m(t) = 10\cos(8\pi\times10^3 t)$

$T_s = 100\mu\text{sec} \quad \rightarrow f_s = 10KHz$

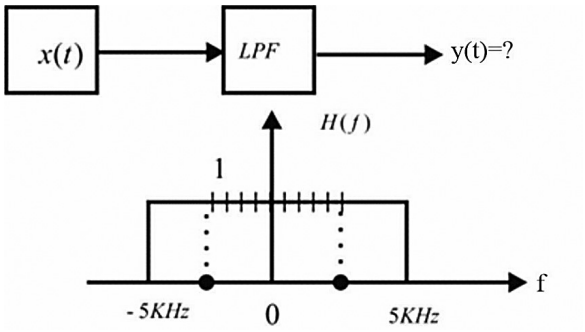


Fig. 8.41

Solution:

$f_m = 4KHz$

$x(t) = p(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s)$

$=$ sampled version of $p(t)$

Method 1:

Ideal setup:

Perfect recovery:

$(f_s > 2f_{max}) \cap (f_m < f_c < f_s - f_m) \cap (PBG = 1/c_o \ in \ |f| \le f_m)$

$(10KHz > 8KHz) \cap (4KHz < 5KHz < 6KHz) \cap (PBG = 1 : |f| \le 4KHz)$

$y(t) = KC_o p(t)$

$y(t) = 1\times\frac{1}{T_s}\times5\times10^{-5}\cos(8\pi\times10^3 t)$

$y(t) = 5\times10^{-5}\times10^4\cos(8\pi\times10^3 t)$

$y(t) = 0.5\cos(8\pi\times10^3 t)$

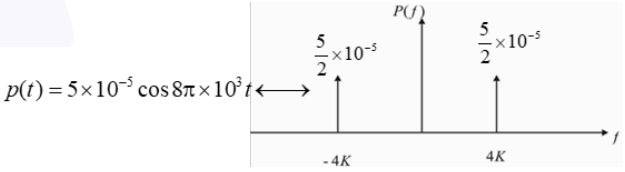
Method 2:

$x(t) = p(t)\sum_{n=-\infty}^{\infty}\delta(t-nT_s) \longleftrightarrow x(f)$

$= \sum_{n=-\infty}^{\infty}C_n P(f-nf_s)$

$f_s = 10^4$

$X(f) = f_s\sum_{n=-\infty}^{\infty}p(f-nf_s)$



$\frac{5}{2}\times10^{-5}\times10^4 = \frac{0.5}{2}$

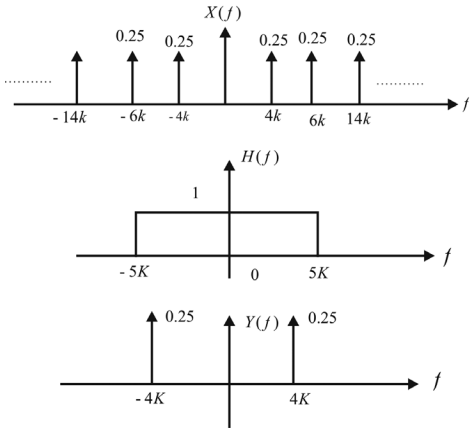


Fig. 8.42

$$Y(f) = 0.25\delta(f-4k) + 0.25\delta(f+4k)$$

$$Y(f) = \frac{1}{5}[0.5\delta(f-4k) + 0.5\delta(f+4k)]$$

$$\cos(2\pi f_0 t) \longleftrightarrow \left\{\frac{1}{2}\delta(f-f_0) + \frac{1}{2}\delta(f+f_0)\right\}$$

$$y(t) = 0.5\cos(2\pi \times 4 \times 10^3 t)$$

3. $x(t) = \sin(14000\pi t)$

$$H(f) = \begin{cases} 1 : |f| \le 12k \\ 0 : |f| > 12k \end{cases}$$

$f_s = 9000$ samples/sec

(i) no of sinusoids in output

(ii) freq at output

(iii) y(t)=?

Solution:

$x(t) = \sin(14000\pi t) \rightarrow f_m = 7KHz$

$\sum \delta(t - nT_s) \rightarrow 0, 9k, 18k, 27k.....$

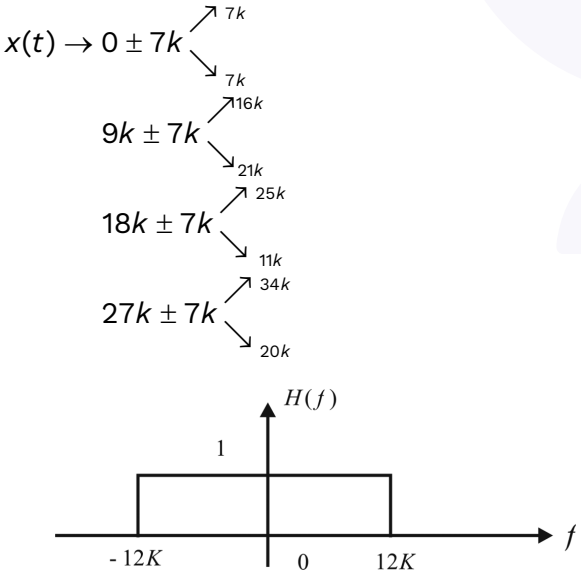


Fig. 8.43

1. 3sinusoids
2. frequency :2k,7k,11k
3. $\underbrace{(f_s > 2f_m)}_{9k>2\times7k\rightarrow fails} \cap (f_m < f_c < f_s - f_m) \cap$

   $(PBG = const \ in \ |f| \le f_m)$

Method 1:

Time domain:

$x_s(t) = x(t).(ct)$

$$x_s(t) = \sin(14000\pi t)\left[a_o + \sum_{n=1}^{\infty} a_n \cos n\omega_s t\right]$$

$f_s = 9000 \qquad \omega_s = 18000\pi$

$$c(t) = \sum \delta(t - nT_s) \longleftrightarrow \begin{cases} c_o = \frac{1}{T_s} \\ c_n = \frac{1}{T_s} \end{cases}$$

$$a_o = \frac{1}{T_s}$$

$$a_n = \frac{2}{T_s}$$

$$x_s(t) = \sin(14000\pi t)[f_s + 2f_s\cos(18000\pi t) + 2f_s\cos[36000\pi t] + ...]$$

$$x_s(t) = \underbrace{f_s \sin(2\pi f_m t)}_{7k} + \overbrace{2f_s \sin(2\pi f_m t)\cos(2\pi f_s t)}^{f_s \pm f_m \nearrow 16k \searrow 2k} + 2f_s \underbrace{\sin(2\pi f_m t)\cos(4\pi f_s t)}$$

$$x_{s_1}(t) = f_s\cos(\omega_n t) - f_s\sin(\omega_s - \omega_n)t - f_s\sin(2\omega_s - \omega_m)t$$

$\omega_s = 18000\pi$

or

$$\sin\omega_n t\begin{bmatrix} f_s + 2f_s\cos\omega_s t + 2f_s\cos 2\omega_s t \\ +2f_s\cos 3\omega_s t \end{bmatrix}$$

$\omega_m = 14000\pi \qquad \omega_s - \omega_m \qquad 2\omega_s - \omega_m \times$

$f_{m=7k} \qquad 2k \qquad 11k$

$$f_s\sin\omega_m t + f_s\sin(\omega_m - \omega_s)t + f_s\sin(\omega_m - 2\omega_s)t$$

$$f_s\sin\omega_m t - f_s\sin(\omega_s - \omega_m)t - f_s\sin(2\omega_s - \omega_m)t$$

$$x_s(t) = 9000[\sin(14000\pi t)] - 9000\sin(4000\pi t) - 9000\sin(22000\pi t)$$

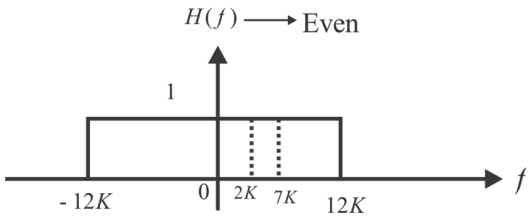


Fig. 8.44

$$y(t) = 9000\sin(14000\pi t)\bullet 1 - 9000\sin(4000\pi t)\bullet 1 - 9000\sin(22000\pi t)$$

Method 2:

Frequency domain:

$$x(t) \longleftrightarrow x(f)$$

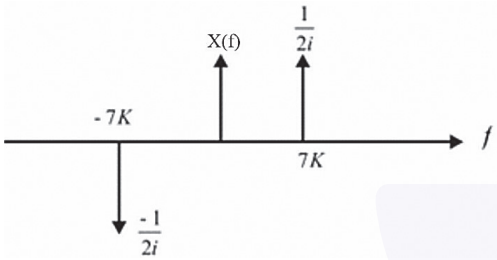


Fig. 8.45

$$x(t)\bullet\sum_{n=-\infty}^{\infty}\delta(t-nT_s) \longleftrightarrow X_s(f) = f_s\sum_{n=-\infty}^{\infty}\times(f-nfs)$$

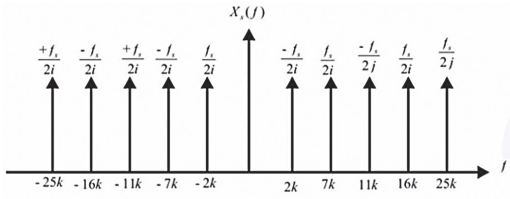


Fig. 8.46

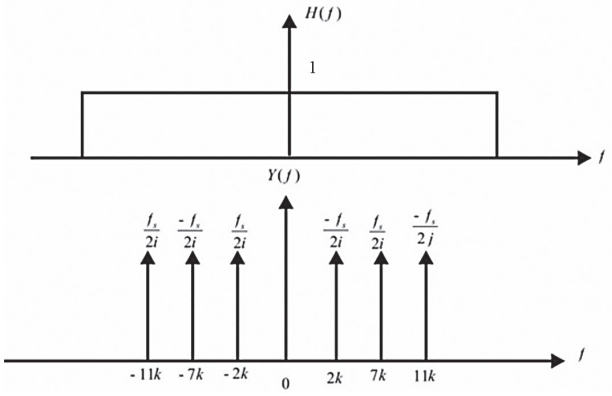


Fig. 8.47

$$Y(f) = \frac{f_s}{2i}[\delta(f-7k) - \delta(f+7k)]$$

$$\frac{-f_s}{2i}[\delta(f-2k) - \delta(f+2k)]$$

$$\frac{-f_s}{2i}[\delta(f-11)k - \delta(f+11k)]$$

$$\downarrow$$

$$y(t) = f_s[\sin(2\pi\times 7\times 10^3 t)] - \sin[2\pi\times 2\times 10^3 t] - \sin[2\pi\times 11\times 10^3 t]$$

$$y(t) = 9000[\sin(14000\pi t) - \sin(4000\pi t) - \sin(22000\pi t)$$

Test Your Understanding

1. A signal $x(t) = 2 + \cos(50\pi t)$ is sampled with sampling interval $T_s = 0.025$ sec and passed through an ideal low-pass filter whose frequency response is shown in the figure

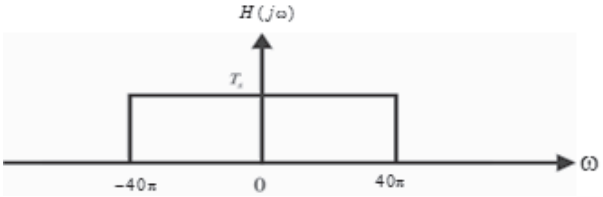


Fig. 8.48

The spectrum of output signal will be

(A)

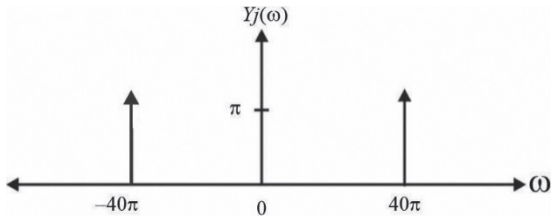


Fig. 8.49

(B)

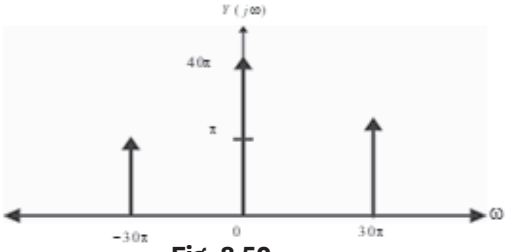


Fig. 8.50

(C)

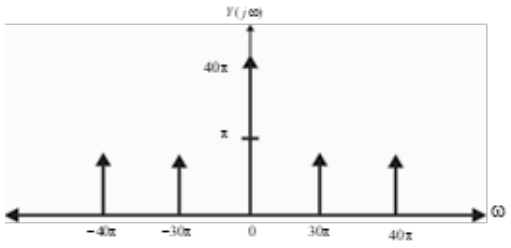


Fig. 8.51

(D) Zero

Solution: (B)

Bandpass Sampling

$m(t)$: Lowpass $\xrightarrow{\text{Bandlimited}}$ f_m

$s(t) = m(t)\cos\omega_c t$: Bandpass signal : $f_c >> f_m$

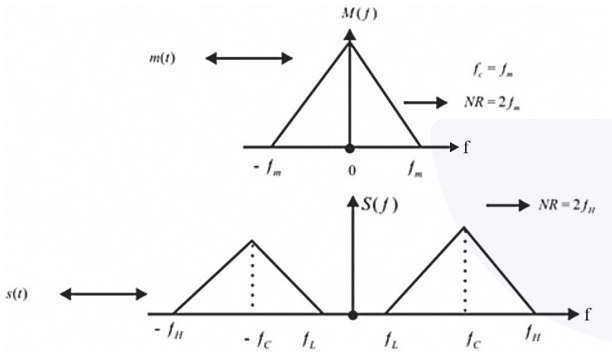


Fig. 8.52

$$f_s \geq \frac{2f_H}{K}$$

$$K = \left[\frac{f_H}{f_H - f_L}\right]$$

$[\bullet] \rightarrow GIF$: Immediate Previous Integer

$$\boxed{\begin{array}{c}[-2.3] = -3 \\ [2] = 2 \\ [2.3] = 2\end{array}}$$

Important Table to Avoid Mistakes

1. First check, what is given? Low pass or band pass sampling. Then decide sampling frequency accordingly.

m(t)	$\delta(t) = m(t)\cos\omega_c t$	f_s of $s(t)$	NR of $s(t)$
Lowpass	Lowpass	$f_s \geq 2f_{max}$ $f_s \geq 2f_H$	$NR = 2f_{max}$ $NR = 2f_H$
Lowpass	Bandpass	$f_s \geq 2f_H / K$ $K = \left[\frac{f_H}{f_H - f_L}\right]$	$NR = 2f_H$

Table 8.4

Test Your Knowledge

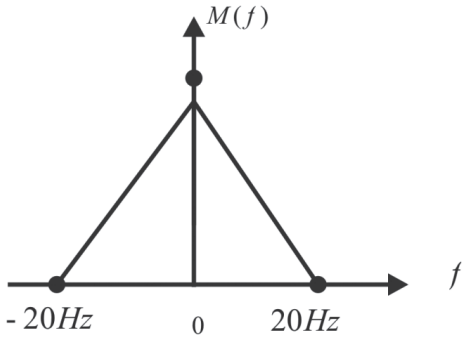


Fig. 8.53

$s(t) = m(t)\cos\left(1000\pi t\right)$

1. $\left(f_s\right)_{min}$ of $s(t)$
2. NR of $s(t)$

Solution:

1. $\left(f_s\right)_{min} = 80$
2. NR = 1040*Hz*

Exercise Level 1

1. In order to recover the original signal from the sampled one, what is the condition to be satisfied for sampling frequency ω_s and highest frequency component ω_m ?

(A) $\omega_m < \omega_s \leq 2\omega_m$

(B) $\omega_s \geq 2\omega_m$

(C) $\omega_s < \omega_m$

(D) $\omega_s = \omega_m$

2. Consider the analog signal $x_a(t) = 3\cos 100\pi t$

The minimum sampling rate F_s required to avoid aliasing will be

(A) 100 Hz
(B) 200 Hz
(C) 300 Hz
(D) 400 Hz

Exercise Level 2

1. A signal represented by $x(t) = 5\cos 400\pi t$ is sampled at a rate 300 samples/s. The resulting samples are passed through an ideal low pass filter of cut-off frequency 150Hz. Which of the following will be contained in the output of the LPF?

(A) 100 Hz

(B) 100 Hz, 150 Hz

(C) 50 Hz, 100 Hz

(D) 50 Hz, 100 Hz, 150 Hz

2. The frequency spectrum of a signal is shown in the figure. If this is ideally sampled at intervals of 1ms, then the frequency spectrum of the sampled signal will be

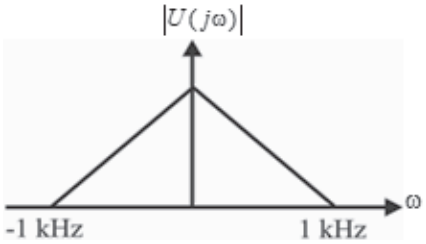


Fig. 8.54

(A)

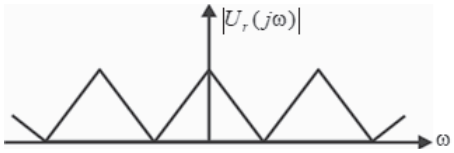


Fig. 8.55

(B)

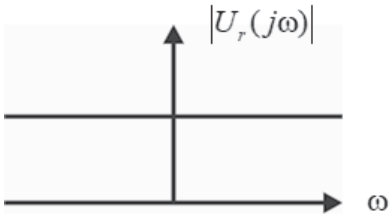


Fig. 8.56

(C)

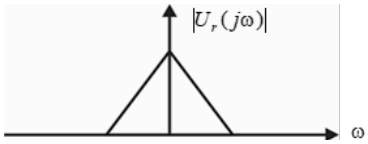


Fig. 8.57

(D)

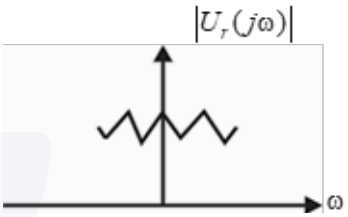


Fig. 8.58

3. An input signal $x(t) = 2 + 5\sin(100\pi t)$ is sampled with a sampling frequency of 400Hz and applied to the system whose transfer function is represented by

$$\frac{Y(z)}{X(z)} = \frac{1}{N}\left(\frac{1-z^{-N}}{1-z^{-1}}\right)$$

Where N represents the number of samples per cycle. The output $y(n)$ of the system under steady state is

(A) 0
(B) 1
(C) 2
(D) 5

Level 1					
1.	B	2.	A		
Level 2					
1.	A	2.	B	3.	C

Table 8.5

Solutions Level 1

1. Solution: (B)

It should satisfy Nyquist criteria

$\omega_s \geq 2\omega_m$

2. Solution: (A)

$x_a(t) = 3\cos 100\pi t$

$f_0 = 50Hz$

$f_{s\ \min} = 2f_0 = 2 \times 50 = 100Hz$

(to avoid aliasing)

Solutions Level 2

1. Solution: (A)

$x(t) = 5\cos 400\pi t$

$f_m = \dfrac{400\pi}{2\pi} = 200Hz$

The input to the LPF will contain frequency components $nf_s \pm f_m, n = 0, 1, 2, \ldots$

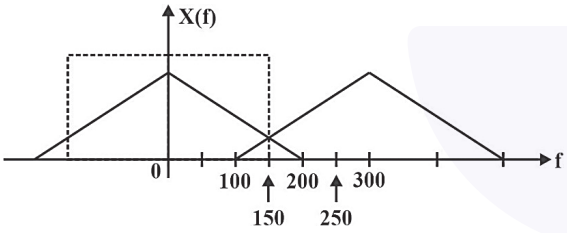


Fig. 8.59

The output of the LPF will contain frequencies which are less than f_c i.e.150Hz.

So $f_s - f_m = 300 - 200 = 100\text{Hz}$ is the only component present in the output of LPF.

2. Solution: (B)

Sampling interval, $T_s = 1\,ms = 10^{-3}$ sec

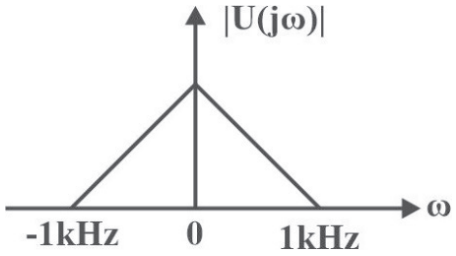


Fig. 8.60

Sampling frequency f_s is given by,

$f_s = \dfrac{1}{T_s} = \dfrac{1}{10^{-3}} = 1kHz$

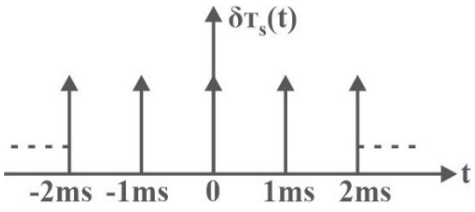


Fig. 8.61

$\delta_{T_s}(t) = \sum_{n=-\infty}^{\infty} \delta\left(t - nT_s\right)$

The sampled signal $u_T(t)$ is expressed as the multiplication of $u(t)$ and $\delta_{T_s}(t)$

$u_T(t) = u(t)\delta_{T_s}(t)$

$u_T(t) = u(t) \sum_{n=-\infty}^{\infty} \delta\left(t - nT_s\right)$

$u_T(t) = \sum_{n=-\infty}^{\infty} u\left(nT_s\right)\delta\left(t - nT_s\right)$

After sampling new signal in the frequency domain is given by,

$U_T(\omega) = \dfrac{1}{T_s} \sum_{n=-\infty}^{\infty} U\left(\omega - n\omega_s\right)$

$U_T(\omega) = \dfrac{1}{T_s}\left[\ldots + U\left(\omega + 2\omega_s\right) + U\left(\omega + \omega_s\right)\right.$

$+U(\omega) + U\left(\omega - \omega_s\right)$

$\left.+U\left(\omega - 2\omega_s\right) + U\left(\omega - 3\omega_s\right) + \ldots.\right]$

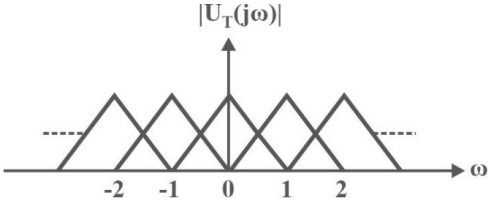
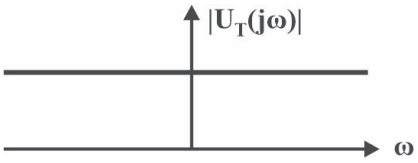


Fig. 8.62

Spectrum of the sampled signal will be,



Hence, the correct option is (B).

3. Solution: (C)

Given: $x(t) = 2 + 5\sin 100\pi t$

Sampling frequency $f_s = 400Hz$

Method 1

Transfer function of a discrete-time system,

$$H(z) = \frac{1}{N}\left[\frac{1 - z^{-N}}{1 - z^{-1}}\right]$$

Where, N = Number of samples taken during every cycle of input x(t)

i.e. $N = \frac{f_s}{f}$, f=Frequency of x(t)

All given conditions can be shown as a block diagram given below,

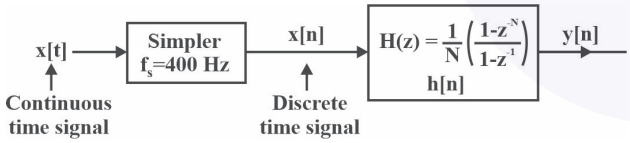


Fig. 8.63

Output of discrete-time system can be obtained as,

$y[n] = x[n] \otimes h[n]$ ...(i)

Given, $x(t) = 2 + 5\sin 100\pi t$

As the combination of a DC and a periodic signal is always periodic with a frequency same as that of the periodic part, so the frequency of x(t)= frequency of $5\sin 100\pi t$.

Frequency of $x(t), \omega = 100\pi \text{rad/sec}$

$$f = \frac{\omega}{2\pi} = 50\text{Hz}$$

From the given condition,

$$N = \frac{f_s}{f} = \frac{400}{50} = 8$$

So, the transfer function of the discrete-time system,

$$H(z) = \frac{1}{8}\left[\frac{1 - z^{-8}}{1 - z^{-1}}\right]$$

$$H(z) = \frac{1}{8}\left[\frac{(1)^2 - \left(z^{-4}\right)^2}{1 - z^{-1}}\right] = \frac{1}{8}\left[\frac{\left(1 - z^{-4}\right)\left(1 + z^{-4}\right)}{\left(1 - z^{-1}\right)}\right]$$

$$H(z) = \frac{1}{8}\left[\frac{\left(1 - z^{-2}\right)\left(1 + z^{-2}\right)\left(1 + z^{-4}\right)}{\left(1 - z^{-1}\right)}\right]$$

$$H(z) = \frac{1}{8}\left[\frac{\left(1 - z^{-1}\right)\left(1 + z^{-1}\right)\left(1 + z^{-2}\right)\left(1 + z^{-4}\right)}{\left(1 - z^{-1}\right)}\right]$$

$$H(z) = \frac{1}{8}\left[\left(1 + z^{-1}\right)\left(1 + z^{-2}\right)\left(1 + z^{-4}\right)\right]$$

$$H(z) = \frac{1}{8}\left[1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6} + z^{-7}\right]$$

Taking inverse z-transform on both sides, the impulse response of the discrete-time system can be obtained as,

$$h[n] = \frac{1}{8}\left\{\begin{matrix}\delta[n] + \delta[n-1] + \delta[n-2] + \delta[n-3] \\ +\delta[n-4] + \delta[n-5] + \delta[n-6] + \delta[n-7]\end{matrix}\right\}$$

From equation (i),

$y[n] = x[n] \otimes h[n]$

$$y[n] = \frac{1}{8}\left\{\begin{matrix}x[n] + x[n-1] + x[n-2] + x[n-3] \\ +x[n-4] + x[n-5] + x[n-6] + x[n-7]\end{matrix}\right.$$

$\{As, x[n] \otimes \delta\left[n \pm n_0\right] = x[n \pm n_0]\}$

$$y[n] = \frac{1}{8}\sum_{K=0}^{7} x[n-K] \quad ...(ii)$$

Where, $x[n] = x(t)\big|_{t=nT_s}$

Given, $f_s = 400\text{Hz}, T_s = \frac{1}{400}\text{sec}$

$$x[n] = [2 + 5\sin 100\pi t]_{t=\frac{\pi}{400}}$$

$$x[n] = 2 + 5\sin\frac{\pi}{4}n$$

Hence, from equation (ii), output y[n]

$$y[n] = \frac{1}{8}\sum_{K=0}^{7} x[n-K]$$

$$y[n]=\frac{1}{8}\sum_{K=0}^{7}\left\{2+5\sin\frac{\pi}{4}(n-K)\right\}$$

$$y[n]=\frac{1}{8}\left[\sum_{K=0}^{7}2+5\sum_{K=0}^{7}\sin\frac{\pi}{4}(n-K)\right]$$

$$y[n]=\frac{1}{8}\left[s_1+5s_2\right] \quad \ldots(iii)$$

Where, $s_1=\sum_{K=0}^{7}2=2\sum_{K=0}^{7}(1)^K=2(7+1)=16$

and $s_2=\sum_{K=0}^{7}\sin\frac{\pi}{4}(n-K)$

$$s_2=\sum_{K=0}^{7}\left(\sin\frac{\pi}{4}n\cos\frac{\pi}{4}K-\cos\frac{\pi}{4}n\sin\frac{\pi}{4}K\right)$$

$$s_2=\sin\frac{\pi}{4}n\sum_{K=0}^{7}\cos\frac{\pi}{4}K-\cos\frac{\pi}{4}n\sum_{K=0}^{7}\sin\frac{\pi}{4}K \quad \ldots(iv)$$

$$\sum_{K=0}^{7}\cos\frac{\pi}{4}K=1+\frac{1}{\sqrt{2}}+0-\frac{1}{\sqrt{2}}-1-\frac{1}{\sqrt{2}}+0+\frac{1}{\sqrt{2}}=0$$

$$\sum_{K=0}^{7}\sin\frac{\pi}{4}K=0+\frac{1}{\sqrt{2}}+1+\frac{1}{\sqrt{2}}+0-\frac{1}{\sqrt{2}}-1-\frac{1}{\sqrt{2}}=0$$

So, from equation (iv),

$$s_2=\left(\sin\frac{\pi}{4}n\right)\times 0-\left(\cos\frac{\pi}{4}n\right)\times 0=0$$

Substituting values of s_1 and s_2 in equation(iii),

$$y[n]=\frac{1}{8}[16+5\times 0]=\frac{16}{8}=2$$

Hence, the steady-state value of output y[n] is 2.

Hence, the correct option is (C).